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Inventor(s)

HWANG; JEONGPIL et al.

HEAT EXCHANGER AND BAFFLE OF THE HEAT EXCHANGER

Abstract

A heat exchanger includes: a shell extending longitudinally; a tube located inside the shell and extending in a longitudinal direction; and a baffle located inside the shell and including a tube through hole through which the tube passes, wherein the baffle includes a fan-shaped plane and sawteeth disposed on at least one side of the fan-shaped plane.

Inventors: HWANG; JEONGPIL (Yongin-si, KR), KIM; Sung-Min (Suwon-si, KR), PARK; Si-Mok (Suwon-si, KR), CHOI; Won-Woo (Suwon-si, KR)

Applicant: SAMSUNG DISPLAY CO., LTD. (YONGIN-SI, KR); Research & Business Foundation SUNGKYUNKWAN UNIVERSITY (Suwon-si, KR)

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

[0001] This application claims priority under 35 U.S.C. § 119 to Korean Patent Application No. 10-2024-0025813 filed at the Korean Intellectual Property Office on Feb. 22, 2024, the disclosure of which is incorporated by reference herein in its entirety.

TECHNICAL FIELD

[0002] The present disclosure relates to heat exchangers and baffles for heat exchangers.

DISCUSSION OF THE RELATED ART

[0003] Generally, a heat exchanger is a device that transfers heat between two or more fluids. Heat exchangers are used in applications related to, for example, heating, cooling, air conditioning, chemical processing, and energy production. Heat can be transferred efficiently through a heat exchanger.

[0004] Shell-and-tube heat exchangers exchange heat using shells and tubes. Heat is transferred between the fluid flowing along the tube and the fluid flowing along the shell around the tube. Shell-and-tube heat exchangers have flexibility and efficiency.

[0005] Typically, baffles may be used in shell-and-tube heat exchangers. Baffles can control the flow direction of fluid flowing along the shell. Additionally, turbulence can be generated in that fluid that flows along the shell.

SUMMARY

[0006] According to an embodiment, a heat exchanger includes: a shell extending longitudinally; a tube located inside the shell and extending in a longitudinal direction; and a baffle located inside the shell and including a tube through hole through which the tube passes, wherein the baffle includes a fan-shaped plane and sawteeth disposed on at least one side of the fan-shaped plane.

[0007] In an embodiment, the sawteeth are bent at a predetermined bending angle.

[0008] In an embodiment, the bending angle is about 20 degrees to about 70 degrees.

[0009] In an embodiment, the bending angles of all of the sawteeth are equal to each other.

[0010] In an embodiment, some of the sawteeth have different bending angles from each other.

[0011] In an embodiment, the baffle is of a plurality of baffles, and baffles adjacent to each other have different vertical directions of the fan-shaped planes than each other.

[0012] In an embodiment, the baffle is of a plurality of baffles, and three consecutive baffles of the plurality of baffles are positioned about 120 degrees apart from each other in a circumferential direction of the shell.

[0013] In an embodiment, a tube is of a plurality of tubes, and a tube through hole is of a plurality of tube through holes.

[0014] In an embodiment, the baffle includes a rod through hole.

[0015] In an embodiment, the heat exchanger further includes: a rod extending through a rod through hole, wherein a radius of the rod through hole is smaller than a radius of the tube through hole.

[0016] In an embodiment, some of the sawteeth do not include at least one tube through hole of the plurality tube through holes.

[0017] In an embodiment, a coating layer is formed on the sawteeth.

[0018] In an embodiment, the coating layer includes at least one of zinc, nickel, chromium, aluminum, or silver.

[0019] In an embodiment, the baffle includes carbon steel.

[0020] In an embodiment, the shape of the sawteeth is trapezoid or triangular.

[0021] According to an embodiment, a baffle for a heat exchanger includes: a fan-shaped plane; a tube through hole through which a tube passes; and sawteeth formed on at least one side of the fan-shaped plane.

[0022] In an embodiment, the sawteeth are bent at a predetermined bending angle.

[0023] In an embodiment, the bending angle is about 20 to about 70 degrees.

[0024] In an embodiment, the bending angles of each of the sawteeth are equal to each other.
[0025] In an embodiment, some of the sawteeth have different bending angles from each other.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0026] FIG. 1 is a diagram schematically illustrating a heat exchanger according to a comparative example.
[0027] FIG. 2 is a cross-sectional view illustrating a heat exchanger according to a comparative example.
[0028] FIG. 3 is a diagram illustrating a baffle according to a comparative example.
[0029] FIG. 4 is a diagram illustrating the shell and baffle of a heat exchanger according to a comparative example.
[0030] FIG. 5 is a diagram schematically illustrating the heat exchanger according to an embodiment.
[0031] FIG. 6 is a diagram illustrating the shell and baffle of the heat exchanger according to an embodiment.
[0032] FIGS. 7, 8, and 9 are diagrams illustrating baffles according to an embodiment.
[0033] FIG. 10 is a diagram illustrating the overlapping of a plurality of baffles according to an embodiment.
[0034] FIG. 11 is a diagram illustrating a baffle according to an embodiment.
[0035] FIG. 12 is an enlarged view of portion A of FIG. 11.
[0036] FIG. 13 and FIG. 14 are views illustrating tubes and baffles of a heat exchanger according to an embodiment.
[0037] FIG. 15 is a diagram illustrating the flow direction of fluid flowing along the shell.
[0038] FIG. 16 is a diagram illustrating simulation results of a comparative example.
[0039] FIG. 17 is a diagram illustrating simulation results of an embodiment.
[0040] FIG. 18 is a view illustrating a coating layer on sawteeth of a baffle according to an embodiment.
[0041] FIG. 19 is a view illustrating some sawteeth that do not include a tube through hole in an embodiment.
[0042] FIG. 20 is a diagram illustrating sawteeth with a trapezoidal shape according to an embodiment.
[0043] FIG. 21 is a view illustrating sawteeth with different shapes according to an embodiment.
[0044] FIG. 22 is a diagram illustrating that the bending angles of a plurality of sawteeth are different from each other according to an embodiment.

DETAILED DESCRIPTION OF THE EMBODIMENTS

[0045] Hereinafter, with reference to the attached drawings, embodiments will be described in detail. The present disclosure may be implemented in many different forms and is not limited to the embodiments described herein.
[0046] Identical or similar components are assigned the same reference numerals throughout the specification and drawings.
[0047] In the drawings, various thicknesses, lengths, and angles are shown and while the arrangement shown does indeed represent an embodiment, it is to be understood that modifications of the various thicknesses, lengths, and angles may be possible within the spirit and scope of the present disclosure and the present disclosure is not necessarily limited to the particular thicknesses, lengths, and angles shown.
[0048] It will be understood that when an element or layer is referred to as being “on” another element or layer, the element or layer may be directly on another element or layer or intervening

elements or layers may be present therebetween. Spatially relative terms, such as “beneath,” “below,” “lower,” “above,” “upper” and the like, may be used herein for ease of description to describe one element or feature's relationship to another element(s) or feature(s) as illustrated in the figures. It will be understood that the spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. For example, if the device in the figures is turned over, elements described as “below” or “beneath” other elements or features would then be oriented “above” the other elements or features. Thus, in the example, terms “below” and “beneath” may encompass both an orientation of above, below and beneath. The device may be otherwise oriented (e.g., rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may be interpreted accordingly.

[0049] A heat exchanger is a device for exchanging heat or energy. Heat exchangers can be used to transfer heat or recover heat. A heat exchanger allows heat to be exchanged between two fluids. Two fluids can exchange heat without being in direct contact. Generally, heat moves from a high-temperature fluid to a low-temperature fluid. Heat exchangers can be used to prevent hot fluids from reaching a certain temperature, or to prevent cold fluids from reaching a certain temperature. In addition, it can be used to increase the efficiency and lifespan of a machine or device. This is because overheated machines or devices may have reduced efficiency and lifespan. Controlling heat movement is very desirable in a variety of industries. Heat exchangers are used in traditional industrial fields such as petrochemical and energy production fields. Recently, they have also been used in the field related to display technology and the semiconductor manufacturing industry.

[0050] Heat exchangers can transfer heat using methods such as convection, conduction, and radiation. Convection can be a method of transferring heat as particles of a fluid move. Conduction can be a way in which heat is transferred within a material or across surfaces that are in contact with one another. Radiation does not require a medium for heat transfer and can be a method of transferring heat by electromagnetic waves. To increase the heat exchange efficiency of the heat exchanger, the heat exchanger is designed to achieve convection, conduction, and radiation methods with increased efficiency.

[0051] Heat exchangers are used in various industrial fields and can be manufactured and used in various types of technology. For example, according to each industrial field, a shell and tube heat exchanger, a plate heat exchanger, a shell and coil heat exchanger, a regenerative heat exchanger, a forced convection heat exchanger, and a radiant heat exchanger may be used.

[0052] Generally, shell-and-tube heat exchangers are a type of heat exchanger that may be used in many industries. Shell-and-tube heat exchangers enable efficient heat transfer. This is because the heat of the fluid flowing along the shell and the fluid flowing along the tube can be transferred directly through the wall of the tube. Additionally, shell-and-tube heat exchangers are generally durable and can have a relatively long service life. Furthermore, shell-and-tube heat exchangers can operate effectively even in high-pressure and high-temperature environments and have the relatively easy maintenance. Generally, a shell and tube heat exchanger includes a shell and tubes. Additionally, the shell and tube heat exchanger may include a baffle.

[0053] Hereinafter, comparative examples of heat exchangers and embodiments will be described with reference to the accompanying drawings.

[0054] FIG. 1 is a diagram schematically illustrating a heat exchanger according to a comparative example.

[0055] The shell sh in FIG. 1 has a cylindrical shape and may extend in the longitudinal direction. FIG. 1 illustrates a cylindrical shell sh, but the present disclosure is not limited thereto. For example, the shell sh may have a square pillar shape, unlike that illustrated in FIG. 1. Additionally, it may have other geometric shapes. That is, the shape of the shell sh is not limited.

[0056] The longitudinal direction, in which the shell sh extends, is the z-axis direction in FIG. 1. When the shell sh is said to have a cylindrical shape, the longitudinal direction may be perpendicular to the circumferential direction of the shell sh. The length to which the shell sh

extends may vary depending on the industrial field to which it is applied. The length of the shell sh must be sufficient for heat exchange, and the longer the length, the more effective it can be for heat exchange.

[0057] When the shell sh has a cylindrical shape, the radius of the cylinder may vary depending on the industrial field to which it is applied. As the cylindrical radius of the shell sh increases, the shell sh of the heat exchanger exc can accommodate more tubes tu. Additionally, it can accommodate more fluid flowing along the shell sh. Accordingly, as the radius of the cylinder of the shell sh increases, heat exchange in the heat exchanger exc may become more efficient. However, as the radius of the cylinder of the shell sh increases, more force may be required to inject the fluid flowing along the shell sh or to sufficiently maintain the speed of the fluid flowing along the shell sh. For example, a larger motor may be needed to introduce fluid, which flows along the shell sh, into the shell sh. Additionally, as the radius of the shell sh increases, the cost for manufacturing the heat exchanger exc increases and the cost for maintaining the heat exchanger exc may increase.

[0058] A tube fluid inlet tu_in and a tube fluid outlet tu_out may be located at both ends of the shell sh in the longitudinal direction. The tube fluid inlet tu_in allows fluid that flows along the tube tu to enter the heat exchanger exc. The tube fluid outlet tu_out allows fluid that flows along the tube tu to exit the heat exchanger exc. The tube fluid inlet tu_in may introduce fluid into the heat exchanger exc by using one tube. A single tube at the tube fluid inlet tu_in may be divided into a plurality of tubes tu inside the heat exchanger exc. This is to expand the contact surface area between the fluid flowing along the tube tu and the fluid flowing along the shell sh. As the surface area increases, heat exchange between fluids may increase, which can increase the efficiency of the heat exchanger exc. The fluid flowing along the tube tu and the fluid flowing along the shell sh do not mix with each other.

[0059] FIG. 1 illustrates the tube fluid inlet tu_in and the tube fluid outlet tu_out on the outer surface of the shell sh. For example, the inside of the heat exchanger exc is not illustrated at the tube fluid inlet tu_in and the tube fluid outlet tu_out. Additionally, FIG. 1 illustrates the inside of a heat exchanger exc where a plurality of tubes tu pass inside the shell sh. This is a cylindrical portion of the shell sh located between the tube fluid inlet tu_in and the tube fluid outlet tu_out, and appears as a longitudinal cross-section of the shell sh of the heat exchanger exc. For example, it appears that a plurality of tubes tu extend in the longitudinal direction inside the shell sh.

[0060] The tube fluid inlet tu_in and the tube fluid outlet tu_out may be cylindrical. For example, each of the tube fluid inlet tu_in and the tube fluid outlet tu_out may have a square pillar shape. Additionally, the tube fluid inlet tu_in and the tube fluid outlet tu_out might not be limited to a particular shape.

[0061] The shell fluid inlet sh_in allows the fluid flowing along the shell sh to enter the heat exchanger exc. The shell fluid outlet sh_out allows the fluid flowing along the shell sh to exit the heat exchanger exc. The fluid flowing along the shell sh that is introduced from the shell fluid inlet sh_in performs heat exchange with the fluid flowing along the tube tu inside the shell sh until the fluid flowing along the shell sh reaches the shell fluid outlet sh_out.

[0062] Each of the shell fluid inlet sh_in and the shell fluid outlet sh_out may be cylindrical in shape. Additionally, each of the shell fluid inlet sh_in and the shell fluid outlet sh_out may have a square pillar shape. Additionally, the shell fluid inlet sh_in and shell fluid outlet sh_out might not be limited to a particular shape.

[0063] The shell fluid inlet sh_in and the shell fluid outlet sh_out may be formed to extend in a direction perpendicular to the direction in which the tube fluid inlet tu_in and the tube fluid outlet tu_out extend. Additionally, the shell fluid inlet sh_in and the shell fluid outlet sh_out may be formed to extend in a direction that is substantially perpendicular to the direction in which the shell sh and the tube tu extend. The shell fluid inlet sh_in may be formed relatively close to the tube fluid outlet tu_out, and the shell fluid outlet sh_out may be formed relatively close to the tube fluid inlet tu_in. This is to increase the heat exchange efficiency of the heat exchanger exc by allowing

the fluid flowing along the shell sh and the fluid flowing along the tube tu to flow in opposite directions of each other.

[0064] Referring to FIG. 1, the tube tu and the shell sh extend in the z-axis direction, and the tube fluid inlet tu_in and the tube fluid outlet tu_out are located at both ends (e.g., opposing ends) of the z-axis of the shell sh. In addition, the shell fluid inlet sh_in and the shell fluid outlet sh_out extend in the y-axis direction, and the shell fluid inlet sh_in is formed adjacent to the tube fluid outlet tu_out. The shell fluid outlet sh_out is formed adjacent to the tube fluid inlet tu_in. The shell sh has a cylindrical shape, and the shell fluid inlet sh_in, the shell fluid outlet sh_out, the tube fluid inlet tu_in, and the tube fluid outlet tu_out also have a cylindrical shape.

[0065] Inside the shell sh, a baffle bf as well as a plurality of tubes tu are located. For example, the baffle bf may include carbon steel. In the heat exchanger exc, the baffle bf can control the flow of fluid flowing along the shell sh and increase heat transfer efficiency. The baffle bf can guide the fluid flowing along the shell sh to follow a certain path. Because of this, the fluid flowing along the shell sh can effectively flow around the plurality of tubes tu. Additionally, the baffle bf can control the speed of fluid flowing along the shell sh. For example, when fluid flows at a speed that is higher than necessary, the baffle can be used as a means to control the speed at which the fluid flows. Additionally, the baffle bf may serve to support the load of the plurality of tubes tu that are located inside the shell sh.

[0066] Additionally, the baffle bf can generate turbulence in the fluid flowing along the shell sh. Turbulence in a fluid can mean that the flow of the fluid is unstable and irregular. The occurrence of turbulence can affect heat transfer efficiency. Turbulence can increase heat transfer efficiency by allowing efficient use of the heat transfer area. Depending on the shape of the baffle bf, turbulence may occur in the fluid flowing along the shell sh, and by appropriately using this, the heat exchange efficiency of the heat exchanger exc can be increased.

[0067] FIG. 1 illustrates a configuration in which a plurality of baffles bf are arranged in the z-axis direction. Additionally, the shell sh and the plurality of tubes tu extend in the longitudinal direction, and the longitudinal direction is the z-axis direction. The shell fluid inlet sh_in extends from the shell sh in the y-axis direction, and the shell fluid outlet sh_out extends from the shell sh in the -y-axis direction. The fluid flowing along the shell sh flows in the z-axis direction. The path of the fluid flowing along the shell sh can be determined by the baffle bf. The fluid flowing along the plurality of tubes tu flows in the z-axis direction. The fluid flowing along the plurality of tubes tu flows into the heat exchanger exc from the tube fluid inlet tu_in through one pipe, and is distributed and flows within the shell sh by a plurality of tubes tu. Afterwards, the plurality of tubes tu are combined again into one tube, and the fluid flows out of the heat exchanger exc through the tube fluid outlet tu_out. In an embodiment, a rod may extend in the longitudinal z-axis direction inside the shell sh. However, the heat exchanger exc may include a rod extending in the z-axis direction.

[0068] FIG. 2 is a cross-sectional view illustrating a heat exchanger according to a comparative example. FIG. 3 is a diagram illustrating a baffle according to a comparative example.

[0069] FIG. 2 is a cross-sectional view of the heat exchanger exc cut in a direction perpendicular to the longitudinal direction in which the shell sh extends. FIG. 2 may be a comparative example that is different from the heat exchanger of FIG. 1. The baffle in FIG. 3 illustrates only one of the three baffles bf illustrated in FIG. 2.

[0070] FIG. 2 illustrates a shell sh and three baffles bf extending in the z-axis direction. The three baffles bf are arranged in the z-axis direction at a predetermined distance from each other. Three baffles bf are located inside the shell sh. The shapes of the three baffles bf are the same as one another. The baffle bf is fan-shaped, and the central angle of the fan shape is 120 degrees. For example, the baffle has a shape of a sector of a circle with a central angle of 120 degrees. Three consecutive baffles are located 120 degrees apart from each other in the circumferential direction of the shell sh. Therefore, when viewed in the z-axis direction, three baffles bf can form one circle.

Since the three baffles bf are 120 degrees apart from each other in the circumferential direction of the shell sh, the fluid flowing along the shell sh can move with a spiral flow. Because of the spiral flow, the heat exchange efficiency of the heat exchanger exc is increase.

[0071] The baffle bf in FIGS. 2 and 3 includes a tube through hole tu_o. There may be multiple tube through holes tu_o. The tube tu extends in the z-axis direction inside the shell sh through the tube through hole tu_o. The plurality of tube through holes tu_o may have the same radius as one another. Additionally, the plurality of tube through holes tu_o may have different radii from each other. The position at which the plurality of tube through holes tu_o are formed in the baffle bf is not limited. Additionally, the number of plurality of tube through holes tu_o is not limited. The radius of the tube through hole tu_o and the radius of the tube tu are substantially the same. Here, the radius of the tube tu includes the thickness of the wall forming the tube tu. When the tube tu is viewed in the z-axis direction, the wall of the tube tu has a circular shape. The wall of the tube tu determines the inside and outside of the tube tu, and the wall of the tube tu allows the fluid flowing along the tube tu to flow into the inside of the tube tu.

[0072] The baffle bf in FIGS. 2 and 3 includes a rod through hole rod_o. The rod extends in the z-axis direction inside the shell sh through the rod through hole rod_o. The rod can ensure that the baffle bf is firmly supported inside the shell sh. Additionally, the rod can prevent the baffle bf from shaking.

[0073] The radius of the rod through hole rod_o may be smaller than the radius of the tube through hole tu_o. Additionally, the radius of the rod through hole rod_o may be larger than the radius of the tube through hole tu_o. Additionally, the radius of the rod through hole rod_o and the radius of the tube through hole tu_o may be the same as each other. However, since the radius of the rod through hole rod_o is smaller than the radius of the tube through hole tu_o, compared to the case where the radius of the rod through hole rod_o is greater than or equal to the radius of the tube through hole tu_o, in the baffle bf, more locations where the tube through hole tu_o may be formed can be secured.

[0074] In the heat exchanger exc using the baffle bf of FIGS. 2 and 3, the plurality of baffles bf do not overlap in the circumferential direction of the shell sh. Therefore, the fluid flowing along the shell sh may leak in the axial direction of the shell sh, which may lead to heat transfer loss.

[0075] FIG. 4 is a diagram illustrating the shell and baffle of a heat exchanger according to a comparative example.

[0076] FIG. 4 is a cross-sectional view of the heat exchanger exc cut in a direction perpendicular to the longitudinal direction in which the shell sh extends. FIG. 4 illustrates a shell sh extending in the z-axis direction and one baffle bf. Baffles bf other than one baffle bf are omitted in FIG. 4.

[0077] The baffle bf in FIG. 4 has a plurality of tube through holes tu_o. Adjacent tube through holes tu_o are separated from each other by the tube through hole distance tu_p. For example, the tube through-hole distance tu_p may be the distance between the centers of adjacent tube through holes tu_o. Assuming that the radius of the tube through holes tu_o are the same, as the tube through hole distance tu_p increases, the number of tube through holes tu_o formed in the baffle bf may decrease. Conversely, as the tube through hole distance tu_p becomes smaller, the number of tube through holes tu_o formed in the baffle bf may increase. As the number of tube through holes tu_o increases, the amount of fluid flowing along the tube tu may increase. However, as the number of tube through holes tu_o increases, the durability of the baffle bf may weaken. This is because most of the area of the baffle bf may be occupied by holes such as the tube through holes tu_o.

[0078] The baffle bf in FIG. 4 has a different shape from the baffle bf in FIGS. 2 and 3. When looking at a cross-section of the heat exchanger exc in the z-axis direction, compared to the case of FIG. 2, the baffle bf of FIG. 4 has a larger surface area than the baffle bf of FIG. 2. Accordingly, three consecutive baffles bf arranged in the z-axis direction overlap each other in the x-y plane even if they are positioned 120 degrees apart from each other in the circumferential direction of the shell sh. Due to this overlap, the heat exchanger exc using the baffle bf of FIG. 4 can prevent fluid

flowing along the shell sh from leaking in the axial direction of the shell sh. Because of this, heat transfer loss can be reduced.

[0079] FIG. 5 is a diagram schematically illustrating the heat exchanger according to an embodiment.

[0080] The shell sh in FIG. 5 has a cylindrical shape and may extend in the longitudinal direction. FIG. 5 illustrates a cylindrical shell sh, but the shell sh may have a square pillar shape, unlike that illustrated in FIG. 5. Additionally, the shell sh may have other geometric shapes. That is, the shape of the shell sh is not limited.

[0081] The longitudinal direction in which the shell sh extends is the z-axis direction in FIG. 5. When the shell sh is said to have a cylindrical shape, the longitudinal direction may be perpendicular to the circumferential direction of the shell sh. The length to which the shell sh extends may vary depending on the industrial field to which it is applied. The length of the shell sh must be sufficient for heat exchange, and the longer the length, the more efficient it can be for heat exchange.

[0082] When the shell sh has a cylindrical shape, the radius of the cylinder may vary depending on the industrial field to which it is applied. As the cylindrical radius of the shell sh increases, the shell sh of the heat exchanger exc can accommodate more tubes tu. Additionally, it can accommodate more fluid flowing along the shell sh. Accordingly, as the radius of the cylinder of the shell sh increases, heat exchange in the heat exchanger exc may become more efficient. However, as the radius of the cylinder of the shell sh increases, more force may be desirable to inject the fluid flowing along the shell sh or to sufficiently maintain the speed of the fluid flowing along the shell sh. For example, a larger motor may be needed to introduce fluid into the shell sh such that the fluid flows along the shell sh. Additionally, as the radius of the shell sh increases, the cost for manufacturing the heat exchanger exc increases and the cost for maintaining the heat exchanger exc may increase.

[0083] A tube fluid inlet tu_in and a tube fluid outlet tu_out may be located at both ends (e.g., opposing ends) of the shell sh in the longitudinal direction. The tube fluid inlet tu_in allows fluid flowing along the tube tu to enter the heat exchanger exc. The tube fluid outlet tu_out allows fluid flowing along the tube tu to exit the heat exchanger exc. In addition, the tube fluid inlet tu_in can introduce fluid into the heat exchanger exc by using one tube. A single tube at the tube fluid inlet tu_in may be divided into a plurality of tubes tu inside the heat exchanger exc. This is to expand the contact surface area between the fluid flowing along a tube tu of the plurality of tubes tu and the fluid flowing along the shell sh. As the surface area increases, heat exchange between fluids becomes more efficient, which can increase the efficiency of the heat exchanger exc. The fluid flowing along the tube tu and the fluid flowing along the shell sh do not mix with each other.

[0084] FIG. 5 illustrates the tube fluid inlet tu_in and the tube fluid outlet tu_out on the outer surface of the shell sh. That is, the inside of the heat exchanger exc is not illustrated at the tube fluid inlet tu_in and tube fluid outlet tu_out. Additionally, FIG. 5 illustrates the inside of a heat exchanger exc where a plurality of tubes tu extend inside the shell sh. This is a cylindrical portion of the shell sh located between the tube fluid inlet tu_in and the tube fluid outlet tu_out, and appears as a longitudinal cross-section of the shell sh of the heat exchanger exc. That is, it appears that a plurality of tubes tu extend in the longitudinal direction inside the shell sh.

[0085] The tube fluid inlet tu_in and the tube fluid outlet tu_out may each have a cylindrical shape. Additionally, each of the tube fluid inlet tu_in and the tube fluid outlet tu_out may have a square pillar shape. Additionally, the tube fluid inlet tu_in and the tube fluid outlet tu_out might not be limited to a particular shape.

[0086] The shell fluid inlet sh_in allows the fluid flowing along the shell sh to enter the heat exchanger exc. The shell fluid outlet sh_out allows the fluid flowing along the shell sh to exit the heat exchanger exc. The fluid, which flows along the shell sh and is introduced from the shell fluid inlet sh_in, may perform heat exchange with the fluid flowing along at least one tube tu of the

plurality of tubes tu inside the shell sh until it reaches the shell fluid outlet sh_out. Each of the shell fluid inlet sh_in and the shell fluid outlet sh_out may have a cylindrical shape. Additionally, each of the shell fluid inlet sh_in and the shell fluid outlet sh_out may have a square pillar shape. Additionally, the shell fluid inlet sh_in and shell fluid outlet sh_out might not be limited to a particular shape.

[0087] The shell fluid inlet sh_in and the shell fluid outlet sh_out may be formed to extend in a direction perpendicular to the direction in which the tube fluid inlet tu_in and the tube fluid outlet tu_out extend. Additionally, the shell fluid inlet sh_in and the shell fluid outlet sh_out may be formed to extend in a direction perpendicular to the direction in which the shell sh and the tube tu extend. The shell fluid inlet sh_in may be formed adjacent to the tube fluid outlet tu_out, and the shell fluid outlet sh_out may be formed adjacent to the tube fluid inlet tu_in. This is to increase the heat exchange efficiency of the heat exchanger exc by allowing the fluid flowing along the shell sh and the fluid flowing along the tube tu to flow in opposite directions of each other.

[0088] Referring to FIG. 5, the tube tu and the shell sh extend in the z-axis direction, and the tube fluid inlet tu_in and the tube fluid outlet tu_out are located at both ends (e.g., opposing ends) of the z-axis of the shell sh. In addition, the shell fluid inlet sh_in and the shell fluid outlet sh_out extend in the y-axis direction. Further, the shell fluid inlet sh_in is formed adjacent to the tube fluid outlet tu_out, and the shell fluid outlet sh_out is formed adjacent to the fluid inlet tu_in. The shell sh has a cylindrical shape, and the shell fluid inlet sh_in, the shell fluid outlet sh_out, the tube fluid inlet tu_in, and the tube fluid outlet tu_out also have a cylindrical shape.

[0089] Inside the shell sh, a baffle bf as well as a plurality of tubes tu are located. For example, the baffle bf may include carbon steel. In the heat exchanger exc, the baffle bf can control the flow of fluid flowing along the shell sh and increase heat transfer efficiency. The baffle bf can guide the fluid that flows along the shell sh to travel a certain path. Because of this, the fluid flowing along the shell sh can effectively flow around the plurality of tubes tu. Additionally, the baffle bf can control the speed of fluid flowing along the shell sh. For example, when fluid flows at a speed that is higher than necessary, the baffle bf can be used as a means to control the speed at which the fluid flows. Additionally, the baffle bf may serve to support the load of the plurality of tubes tu that are located inside the shell sh.

[0090] Additionally, the baffle bf can generate turbulence in the fluid flowing along the shell sh. Turbulence in a fluid can mean that the flow of the fluid is unstable and irregular. The occurrence of turbulence can affect heat transfer efficiency. Turbulence can increase heat transfer efficiency by allowing efficient use of the heat transfer area. Depending on the shape of the baffle bf, turbulence may occur in the fluid flowing along the shell sh, and by appropriately using this, the heat exchange efficiency of the heat exchanger exc can be increased.

[0091] FIG. 5 illustrates a configuration in which a plurality of baffles bf are arranged in the z-axis direction. Additionally, the shell sh and the plurality of tubes tu extend in the longitudinal direction, and the longitudinal direction is the z-axis direction. The baffle bf includes a fan-shaped plane fp. For example, the baffle bf may be a segment or portion of a circle. A fan-shaped plane fp is formed with one curved line and two straight lines as edges. The two straight lines can each have the same length as each other. Additionally, the two straight lines may have different lengths from each other. For example, one curve of the fan-shaped plane fp might not be limited to a particular curvature. One curve of the fan-shaped plane fp can be formed corresponding to the shape of the shell sh. The direction in which the fan-shaped plane fp of the baffle bf extends may be different for each baffle bf. The shell fluid inlet sh_in extends from the shell sh in the y-axis direction, and the shell fluid outlet sh_out extends from the shell sh in the y-axis direction. The fluid flowing along the shell sh flows in the z-axis direction. The path of the fluid flowing along the shell sh can be determined by the baffle bf. The fluid flowing along the tube tu flows in the z-axis direction. The fluid flowing along the tube tu flows into the heat exchanger exc from the tube fluid inlet tu_in through one pipe, and is distributed and flows within the shell sh by a plurality of tubes tu. Afterwards, the plurality

of tubes tu are combined again into one tube, and the fluid flows out of the heat exchanger exc through the tube fluid outlet tu_out. In an embodiment, a rod may extend in the longitudinal z-axis direction inside the shell sh. For example, the heat exchanger exc may include a rod extending in the z-axis direction.

[0092] That is, compared to the heat exchanger exc of FIG. 1, the shape and posture of the baffle bf of the heat exchanger exc of FIG. 5 are different from each other.

[0093] Below, the shape of the baffle bf according to an embodiment will be described.

[0094] FIG. 6 is a diagram illustrating the shell and baffle of the heat exchanger according to an embodiment.

[0095] The baffle bf in FIG. 6 includes a fan-shaped plane fp, which includes a tube through-hole tu_o. The tube tu penetrates the baffle bf through the tube through hole tu_o. In FIG. 6, the fan-shaped plane fp of the baffle bf is formed with curves and straight lines. In FIG. 6, the fan-shaped plane fp of the baffle bf is formed in the x-y plane. In FIG. 6, the traveling direction of the fan-shaped plane fp of the baffle bf may be in the x-y plane. For example, in FIG. 6, the direction in which the fan-shaped plane fp of the baffle bf moves is perpendicular to the z-axis direction. The fan-shaped plane fp of the baffle bf extends in the x-y plane.

[0096] According to an embodiment, at least one side of the fan-shaped plane fp of the baffle bf includes a sawtooth. One side of the fan-shaped plane fp may be an end edge of the baffle bf. The sawtooth may be plural. In FIG. 6, the sawteeth include as a first sawtooth bf_t1, a second sawtooth bf_t2, and a third sawtooth bf_t3. In FIG. 6, the shape of each sawtooth is triangular. However, the shape of the sawtooth is not limited. For example, the shape of each sawtooth may be trapezoidal.

[0097] Since the baffle bf of the heat exchanger exc includes the sawtooth on one side of the fan-shaped plane fp, a strong turbulence intensity can be obtained in the fluid flowing along the shell sh. Like the baffle bf in FIG. 4, a baffle bf that does not include a sawtooth on one side of the fan-shaped plane fp may have difficulty obtaining strong turbulence in the fluid flowing along the shell sh. The fluid flowing along the shell sh in the heat exchanger exc proceeds along a spiral flow path. If one side of the baffle bf does not include the sawtooth and is flat, the ability to disturb the spiral flow of the fluid flowing along the shell sh may be weak. This may mean that no active heat exchange occurs in the heat exchanger exc. An embodiment includes the sawtooth on at least one side of the fan-shaped plane fp of the baffle bf. For example, the baffle bf may include the sawtooth on one side of the fan-shaped plane fp in a direction towards a center of the shell sh. For example, the side of the baffle bf including the sawtooth may face inward of the shell sh, and the curved side of the baffle bf may face an inner sidewall of the shell sh. Additionally, an embodiment may include the sawtooth at the end edge of the baffle bf. The sawtooth shape of the baffle bf may disturb the spiral flow of fluid flowing along the shell sh. Accordingly, the intensity of turbulence may become stronger, and active heat exchange can occur in the heat exchanger exc.

[0098] One side of the fan-shaped plane fp, in which the sawtooth is formed, may be a portion where the baffle bf and another baffle bf overlap each other. Due to this overlap, it is possible to prevent the fluid flowing along the shell sh from leaking in the axial direction of the shell sh, thereby reducing the amount of fluid flow. This can result in minimizing heat transfer loss.

[0099] FIG. 7 to FIG. 9 are diagrams illustrating baffles according to an embodiment.

[0100] FIG. 7 to FIG. 9 illustrate the same baffle bf. The baffle bf of FIGS. 7 to 9 includes sawteeth on one side of the fan-shaped plane fp in the direction in which it moves, like the baffle bf of FIG. 6. Furthermore, the sawteeth are bent at a predetermined bending angle based on the direction of travel of the fan-shaped plane fp. That is, in FIGS. 7 to 9, the sawteeth are bent in the-z-axis direction.

[0101] FIG. 7 is a view looking at the baffle bf in the z-axis direction. The baffle bf appears in the x-y plane. The baffle bf includes a tube through hole tu_o. There may be multiple tube through holes tu_o. The baffle bf in the embodiment of FIG. 7 may also include the rod through hole rod_o.

[0102] The plurality of sawteeth of the baffle bf in FIG. 7 include a first sawtooth bf_t1, a second

sawtooth bf_t2, and a third sawtooth bf_t3. For example, the baffle bf in FIG. 7 includes three sawteeth. However, in the present embodiment, there is no limitation on the number of sawteeth, and for example, two or four or more sawteeth may be formed on the baffle bf.

[0103] A tube through hole tu_o is formed in the first sawtooth bf_t1 and the second sawtooth bf_t2 in FIG. 7. However, there is no limit to the position where the tube through hole tu_o is formed. For example, tube through-holes tu_o might not be formed in the sawteeth. For example, tube through-holes tu_o may be formed in all of the sawteeth.

[0104] FIG. 8 is a view looking at the baffle bf in the x-axis direction. The baffle bf appears in the z-y plane. FIG. 9 is a view looking at the baffle bf in the x-axis direction. The baffle bf appears in the z-y plane. FIG. 8 and FIG. 9 illustrate the curved shape of the sawteeth more clearly.

[0105] FIG. 10 is a diagram illustrating the overlapping shape of a plurality of baffles according to an embodiment.

[0106] The baffle bf in FIG. 10 has the same shape as that of the baffle bf in FIGS. 7 to 9. FIG. 10 illustrates a shape in which three baffles bf overlap in the x-y plane. By the three baffles bf partially overlapping each other, the fluid flowing along the shell sh in the heat exchanger exc can be prevented from leaking in the axial direction of the shell sh. Because of this, heat transfer loss can be reduced.

[0107] In FIG. 7 to FIG. 10, when the sawteeth included in one side of the baffle bf are bent in the z-axis direction, the effect of disturbing the spiral flow of fluid flowing along the shell sh can be increased. For example, by applying the bending angle to the sawtooth shape, the increase in turbulence intensity of the fluid flowing along the shell sh can be maximized. This allows the heat exchanger exc to have higher heat transfer efficiency, since it is possible to enlarge the contact area with the tube tu wall and the residence time of the fluid.

[0108] FIG. 11 is a diagram illustrating a baffle according to an embodiment.

[0109] FIG. 11 is a perspective view illustrating a baffle bf whose sawteeth are bent at a predetermined angle. The baffle bf in FIG. 11 is the same as the baffle bf in FIGS. 7 to 10. Part A of FIG. 11 will be enlarged to describe the baffle bf of the embodiment in more detail.

[0110] FIG. 12 is an enlarged view of portion A of FIG. 11.

[0111] FIG. 12 illustrates the fan-shaped plane fp and sawteeth of the baffle bf. The tube through hole tu_o is formed in the fan-shaped plane fp, the first sawtooth bf_t1, and the second sawtooth bf_t2. There is no limit to the position where the tube through hole tu_o is formed in the baffle bf. For example, the tube through hole tu_o might not be formed in the first sawtooth bf_t1 and the second sawtooth bf_t2, and the tube through hole tu_o may be formed in the third sawtooth bf_t3.

[0112] In FIG. 12, the first sawtooth bf_t1, the second sawtooth bf_t2, and the third sawtooth bf_t3 are bent by the bending angle a. For example, each sawtooth is bent by a bending angle a based on the direction in which the fan-shaped plane fp of the baffle bf extends. The bending angle a can exceed 0 degrees and can be 180 degrees. The fact that the sawtooth may be bent by 180 degrees may mean that the sawtooth and the fan-shaped plane fp are in contact with each other. The reason why the sawteeth have a bending angle a is to maximize the increase in turbulence intensity of the fluid flowing along the shell sh. Depending on the numerical range of the bending angle a, the amount of increase in turbulence intensity of the fluid flowing along the shell sh may vary. For example, the bending angle a may be between about 20 and about 70 degrees.

[0113] FIG. 12 illustrates three sawteeth: a first sawtooth bf_t1, a second sawtooth bf_t2, and a third sawtooth bf_t3, but the present disclosure is not limited thereto. For example, the baffle bf may have two or four or more sawteeth. As the number of sawteeth changes, the shape of the sawteeth may also change. As the number of sawteeth increases, the size of each sawtooth may become smaller. The number of sawteeth formed on the baffle bf can be determined so that the fluid flowing along the shell sh has optimal turbulence strength.

[0114] Additionally, FIG. 12 illustrates that the sawteeth bf_t1, bf_t2, and bf_t3 provided in the baffle bf are triangular in shape. However, in the present disclosure, the shape of the sawteeth

bf_t1, bf_t2, and bf_t3 is not limited. For example, the sawteeth bf_t1, bf_t2, and bf_t3 may be trapezoidal in shape. Additionally, the sawteeth bf_t1, bf_t2, and bf_t3 may have a semicircular shape. The shape of the sawteeth can be determined so that the fluid flowing along the shell sh has optimal turbulence intensity.

[0115] Below, the posture of the baffle bf in according to an embodiment will be described.

[0116] The posture of the baffle bf may mean the tilt angle or rotation angle that the baffle bf has with respect to the tube tu and the shell sh. Additionally, the posture of the baffle bf may be the angle difference between the direction in which the fan-shaped plane fp of the baffle bf extends and the direction in which the tube tu travels. Additionally, the posture of the baffle bf may be the angle difference between the vertical direction of the fan-shaped plane fp of the baffle bf and the traveling direction of the tube tu.

[0117] FIG. 13 and FIG. 14 are views illustrating tubes and baffles of a heat exchanger according to an embodiment. FIG. 13 illustrates a shape in which a plurality of tubes tu penetrate three baffles bf. Three consecutively arranged baffles bf are each located in different parts of the x-y plane. When viewed in the z-axis direction, three consecutively arranged baffles bf form a circle, and may partially overlap each other. The flow of fluid along the shell sh due to the three sequentially arranged baffles bf has a spiral flow.

[0118] The arrow at the end of the tube tu indicates the direction of fluid flowing along the tube tu. In the case of FIG. 13, the fluid flowing along the tube tu flows in the z-axis direction. For example, the fluid flowing along the shell sh flows in the z-axis direction for high heat exchange efficiency. However, it is possible for the fluid flowing along the shell sh to flow in the z-axis direction like the fluid flowing along the tube tu.

[0119] The three baffles bf in FIG. 13 are aligned along the z-axis direction. The heat exchanger exc includes a first baffle bf1, a second baffle bf2, and a third baffle bf3 along the z-axis direction. Each baffle bf has the same shape as one another. Additionally, each baffle bf has sawteeth having a bending angle a. In FIG. 13, the sawteeth of each baffle bf are bent at a predetermined bending angle based on the advancing direction of the fan-shaped plane fp of each baffle bf. For example, the direction in which the sawteeth are bent may be opposite to the direction in which the fluid flowing along the shell sh moves; however, the present disclosure is not limited thereto, and for example, the direction in which the sawteeth are bent may be the same as the direction in which the fluid flowing along the shell sh moves. Accordingly, it is possible to maximize the turbulence intensity of the fluid flowing along the shell sh.

[0120] FIG. 14 illustrates three consecutive baffles bf and a plurality of tubes tu in the heat exchanger exc. FIG. 14 illustrates the shape of three baffles bf and a plurality of tubes tu when viewed in the x-axis direction. In order along the z-axis direction, the three baffles bf are the first baffle bf1, the second baffle bf2, and the third baffle bf3. Each baffle bf may have an inclined posture with respect to the y-axis.

[0121] In other words, each baffle bf may extend at a predetermined angle with the y-axis direction. Additionally, the fan-shaped plane fp of each baffle bf may extend at a predetermined angle with respect to the y-axis direction. Additionally, each baffle bf may have a posture rotated in the circumferential direction based on the x-axis direction. The baffle bf of the comparative example in FIG. 1 extends in the y-axis direction. Additionally, the fan-shaped plane fp of the baffle bf of the comparative example in FIG. 1 extends in the y-axis direction.

[0122] Each baffle bf may have an inclined posture with respect to the x-axis. In other words, each baffle bf may extend at a predetermined angle with respect to the x-axis direction. For example, each baffle bf may be positioned to be slanted with respect to the x-axis direction. Additionally, the fan-shaped plane fp of each baffle bf may extend at a predetermined angle with respect to the x-axis direction. Additionally, each baffle bf may have a posture that is rotated in the circumferential direction based on the y-axis direction. The baffle bf of the comparative example in FIG. 1 extends in the x-axis direction. Additionally, the fan-shaped plane fp of the baffle bf of the comparative

example in FIG. 1 extends in the x-axis direction.

[0123] Each baffle bf arranged in succession may have a different posture. Additionally, baffles bf that are adjacent to each other may have different vertical directions of the fan-shaped planes fp. For example, the first baffle bf1 may have a tilt of about 20 degrees in relation to the y-axis, and a tilt of 0 degrees in relation to the x-axis. For example, the second baffle bf2 may have a tilt of about 30 degrees in relation to the y-axis, and a tilt of about 20 degrees in relation to the x-axis. In addition, when the first baffle bf1 has a tilt of about 40 degrees in the y-axis direction and a tilt of about 20 degrees in the x-axis direction, and the second baffle bf2 can have a tilt of 0 degrees in the y-axis direction and a tilt of about 20 degrees in the x-axis direction.

[0124] Additionally, each baffle bf arranged in succession may have the same posture as one another. For example, when the first baffle bf1 has a tilt of about 20 degrees in relation to the y-axis, and a tilt of 0 degrees in relation to the x-axis, the second baffle bf2 and the third baffle bf3 can also have a tilt of about 20 degrees in relation to the y-axis, and a tilt of 0 degrees in relation to the x-axis.

[0125] In this way, the posture of each baffle bf may be different from each other or may be the same as each other. Additionally, the postures of some baffles bf may be the same as each other while the postures of other baffles bf may be different from each other. The turbulence intensity of the fluid flowing along the shell sh may vary depending on the posture of the baffle bf. The attitude of the baffle bf may be determined so that the fluid flowing along the shell sh has a desirable turbulence intensity.

[0126] FIG. 15 is a diagram illustrating the flow direction of fluid flowing along the shell. FIG. 15 illustrates that the fluid flowing along the shell sh has a spiral flow due to three baffles bf arranged in succession in the z-axis direction. The arrows in FIG. 15 indicate a spiral flow of fluid flowing along the shell sh.

[0127] The spiral flow of fluid flowing along the shell sh will be explained. When looking at the three baffles bf in the z-axis direction, the y-axis direction in the drawing may be referred to as the 12 o'clock direction, and the x-axis direction may be referred to as the 3 o'clock direction. The fluid flowing along the shell sh flows from approximately 7 o'clock to approximately 1 o'clock when flowing from the first baffle bf1 to the second baffle bf2. The fluid flowing along the shell sh flows from approximately 1 o'clock to approximately 5 o'clock when flowing from the second baffle bf2 to the third baffle bf3. After the third baffle bf3, the fluid flowing along the shell sh may flow from approximately 5 o'clock to approximately 7 o'clock. Accordingly, the fluid flowing along the shell sh may have a clockwise spiral flow while moving in the z-axis direction. Of course, depending on the shape and posture of the baffle bf, the fluid flowing along the shell sh may have a counterclockwise spiral flow. For example, heat efficiency may be increased for heat exchange if the sawteeth of the baffle bf are bent in a direction opposite to the direction of flow of the fluid flowing along the shell sh.

[0128] FIG. 16 is a diagram illustrating simulation results of a comparative example. FIG. 17 is a diagram illustrating simulation results of an embodiment.

[0129] FIGS. 16 and 17 illustrate turbulence intensity according to color according to the position of the heat exchanger exc. FIG. 16 and FIG. 17 illustrate a tube tu and a baffle bf. The fluid flowing along the shell sh flows in the z-axis direction. The arrow illustrated in the upper left corner of FIGS. 16 and 17 indicates the direction of fluid flowing along the shell sh. The comparative example of FIG. 16 and the embodiment of FIG. 17 have the baffle bf in the same posture. However, the baffle bf of the comparative example of FIG. 16 does not have sawteeth, but the baffle bf of the example of FIG. 17 has sawteeth formed with a bending angle a.

[0130] Looking at FIG. 17, it can be seen that the intensity of turbulence increases at the location where the fluid, which flows along the shell sh, comes into contact with the baffle bf. This indicates that strong turbulence occurred in the fluid flowing along the shell sh due to the sawteeth of the baffle bf. As strong turbulence occurs in the fluid flowing along the shell sh, the heat exchange

efficiency of the heat exchanger exc may increase.

[0131] FIG. **18** is a view illustrating a coating layer on the sawteeth of the baffle according to an embodiment.

[0132] Because the sawteeth of the baffle bf create strong turbulence in the fluid flowing along the shell sh, it may be more desirable for the baffle bf with sawteeth to have corrosion resistance than the baffle bf without sawteeth. Therefore, corrosion resistance can be increased by forming a coating layer bf_c on the sawteeth of the baffle bf. The coating layer bf_c may be an anti-corrosion coating. For example, the coating layer bf_c may include at least one of zinc (Zn), nickel (Ni), chromium (Cr), aluminum (Al), and/or silver (Ag). By forming a coating layer bf_c on the sawteeth, corrosion of the sawteeth can be prevented. Additionally, the durability of the baffle bf can be increased.

[0133] FIG. **19** is a view illustrating some sawteeth that do not include a tube through hole according to an embodiment.

[0134] In FIG. **19**, the tube through hole tu_o is not formed in the first sawtooth bf_t1 and the third sawtooth bf_t3. Accordingly, the first sawtooth bf_t1 and the third sawtooth bf_t3 can have a bending angle a more freely than the second sawtooth bf_t2. Since the second sawtooth bf_t2 includes the tube through hole tu_o, the bending angle a is limited. This is because, if the second sawtooth bf_t2 is excessively bent, interference may occur between the end of the second sawtooth bf_t2 and the tube tu. Additionally, if the second sawtooth bf_t2 is excessively bent, it might not have a surface sufficient to form the tube through hole tu_o. However, in cases where the tube through hole tu_o is not formed in the sawtooth, such as the first sawtooth bf_t1 and the third sawtooth bf_t3, the bending angle of the sawtooth is not limited by a tube tu traveling through a tube through hole tu_o. Accordingly, there may be no limitation in forming an appropriate bending angle in the sawteeth to form a strong turbulent strength in the fluid that flows along the shell sh.

[0135] FIG. **20** is a diagram illustrating sawteeth with a trapezoidal shape according to an embodiment.

[0136] In this way, the sawteeth bf_t1, bf_t2, and bf_t3 might not only be triangular but also trapezoidal in shape. Of course, shapes other than triangles and trapezoids are also possible. In other words, there is no limitation with regard to the shape of the sawteeth bf_t1, bf_t2, and bf_t3. The shape of the sawteeth bf_t1, bf_t2, and bf_t3 may be determined so that the fluid flowing along the shell sh has increased turbulence strength.

[0137] FIG. **21** is a view illustrating sawteeth with different shapes according to an embodiment.

[0138] FIG. **21** illustrates that the baffle bf has four sawteeth. The number of sawteeth of the baffle bf is not limited. Accordingly, the baffle bf may have 1, 2, 3, or 4 or more sawteeth. Additionally, the sawteeth of the baffle bf may be of different sizes from each other. Referring to FIG. **21**, the sizes of the first sawtooth bf_t1 and the second sawtooth bf_t2 are the same as each other.

However, for example, sizes of the first sawtooth bf_t1 and the second sawtooth bf_t2, a size of the third sawtooth bf_t3, and a size of the fourth sawtooth bf_t4 are different from each other. The size of the sawtooth of the baffle bf is not limited. The number and size of sawteeth can be determined so that the fluid flowing along the shell sh has increased turbulence strength.

[0139] FIG. **22** is a diagram illustrating that the bending angles of a plurality of sawteeth are different from each other according to an embodiment.

[0140] Referring to FIG. **22**, the first sawtooth bf_t1 has a bending angle b, but each of the second sawtooth bf_t2 and the third sawtooth bf_t3 do not have a bending angle. The second sawtooth bf_t2 and the third sawtooth bf_t3 are continuous in the direction of travel of the fan-shaped plane fp. In this way, each sawtooth bf_t1, bf_t2, and bf_t3 may have a different bending angle. Because of this, the degree of freedom is increased in determining that the fluid flowing along the shell sh has a desirable turbulence intensity. The difference in the bending angle of each sawtooth can be determined so that the fluid flowing along the shell sh has increased turbulence strength.

[0141] While the present disclosure has been described with reference to embodiments thereof, it

will be understood by those of ordinary skill in the art that various changes in form and details may be made thereto without departing from the spirit and scope of the present disclosure.

Claims

1. A heat exchanger, comprising: a shell extending longitudinally; a tube located inside the shell and extending in a longitudinal direction; and a baffle located inside the shell and including a tube through hole through which the tube passes, wherein the baffle includes a fan-shaped plane and sawteeth disposed on at least one side of the fan-shaped plane.
 2. The heat exchanger of claim 1, wherein the sawteeth are bent at a predetermined bending angle.
 3. The heat exchanger of claim 2, wherein the bending angle is about 20 degrees to about 70 degrees.
 4. The heat exchanger of claim 2, wherein the bending angles of all of the sawteeth are equal to each other.
 5. The heat exchanger of claim 2, wherein some of the sawteeth have different bending angles from each other.
 6. The heat exchanger of claim 1, wherein the baffle is of a plurality of baffles, and baffles adjacent to each other have different vertical directions of the fan-shaped planes than each other.
 7. The heat exchanger of claim 1, wherein the baffle is of a plurality of baffles, and three consecutive baffles of the plurality of baffles are positioned about 120 degrees apart from each other in a circumferential direction of the shell.
 8. The heat exchanger of claim 1, wherein a tube is of a plurality of tubes, and a tube through hole is of a plurality of tube through holes.
 9. The heat exchanger of claim 8, wherein the baffle includes a rod through hole.
 10. The heat exchanger of claim 9, further comprising: a rod extending through a rod through hole, wherein a radius of the rod through hole is smaller than a radius of the tube through hole.
 11. The heat exchanger of claim 8, wherein some of the sawteeth do not include at least one tube through hole of the plurality tube through holes.
 12. The heat exchanger of claim 1, wherein a coating layer is formed on the sawteeth.
 13. The heat exchanger of claim 12, wherein the coating layer includes at least one of zinc, nickel, chromium, aluminum, or silver.
 14. The heat exchanger of claim 1, wherein the baffle comprises carbon steel.
 15. The heat exchanger of claim 1, wherein the shape of the sawteeth is trapezoid or triangular.
 16. A baffle for a heat exchanger, comprising: a fan-shaped plane; a tube through hole through which a tube passes; and sawteeth formed on at least one side of the fan-shaped plane.
 17. The baffle for the heat exchanger of claim 16, wherein the sawteeth are bent at a predetermined bending angle.
 18. The baffle for the heat exchanger of claim 17, wherein the bending angle is about 20 to about 70 degrees.
 19. The baffle for the heat exchanger of claim 17, wherein the bending angles of each of the sawteeth are equal to each other.
 20. The baffle for the heat exchanger of claim 17, wherein some of the sawteeth have different bending angles from each other.
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